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Structural features of a Eu^{3+} doped nuclear glass and gels obtained from glass leaching

N. Ollier ^{a,*}, G. Concas ^b, G. Panczer ^a, B. Champagnon ^a, T. Charpentier ^c

^a *Laboratoire de Physico-Chimie des Matériaux Luminescents, UMR 5620 CNRS, Université Claude Bernard, 69622 Villeurbanne, France*

^b *Dipartimento di Fisica, Università di Cagliari, and Istituto Nazionale per la Fisica della Materia, SP Monserrato-Sestu Km 0.700, I-09042 Monserrato (Cagliari), Italy*

^c *Service de Chimie Moléculaire – CEA Saclay – 91191 Gif-sur-Yvette, cedex, France*

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Abstract

In order to describe the rare earth environment in an aluminoborosilicate nuclear glass and its evolution with water leaching, Eu^{3+} was used as a luminescent structural probe. Two types of Eu^{3+} environments were observed in the glass as well as in the gels (originating from leached glass). A study of the B_2 crystal field parameters allows one to identify one site as a silicate type. We showed moreover that the second site is modified with leaching from a borate to an aluminate site. Asymmetry ratio from photoluminescence and Mössbauer analyses demonstrated a decrease of the average distortion of the Eu^{3+} sites in the gels. Indeed, gel can be described as an amorphous matrix but less disordered than glass. Another important result is that the lanthanide environment is not influenced by the leaching mode.

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1. Introduction

The SON68 glass is the French inactive reference for the light water reactor nuclear waste storage. A lot of studies have described the glass leaching processes with the apparition of an aluminosilica layer in surface generally called ‘gel’. The gel is depleted of most alkali ions and boron and is enriched in Al, Zr, Zn and rare earth element (REE) [1–4]. Structural properties of this

layer are of great importance in predicting the long-term behaviour of nuclear glasses. In particular, REE are good actinide analogues and are found as fission products in active glasses. Although REE retention is known to be high [2], a very limited number of studies on REE structural environment in nuclear glass and gel have been performed up to now.

In contrast, a lot of REE doped glassy matrices have been investigated by fluorescence line narrowing (FLN) as potential laser materials [5–9]. As a matter of fact, site selective spectroscopy using tunable lasers allows one to reach subsets of ions in the glassy matrix and to describe the local structure around the rare earth element. Furthermore, structural changes in microscopic environment of

* Corresponding author. Present address: Domaine Scientifique de la DOUA, Bat. Lippmann, 12 rue Ampère 43 Bd du 11 Novembre, 69622 Villeurbanne, cedex, France. Tel.: +33-1 69 33 45 18; fax: +33-1 69 33 30 22/4 72 43 11 30.

E-mail address: ollier@hplsesi.polytechnique.fr (N. Ollier).

glasses (phase transition or pressure) can be detected using FLN studies [9]. In the present study, Eu^{3+} was used as a structural probe to follow the rare earth behaviour after leaching. These experiments were completed by ^{151}Eu Mössbauer spectroscopy and ^{27}Al solid state NMR studies.

2. Experimental procedure

The aluminoborosilicate glass was synthesized at 1200 °C in a Pt crucible and annealed 1 h at 520 °C in a graphite crucible. Its composition is close to the French nuclear glass SON68 [1] doped with 0.81% of Eu (instead of Nd and Pr) and is given in Table 1. Two complete leaching were realized in a Teflon reactor at 90 °C, in a pseudo-dynamic mode. Glass powder was altered in order to improve the glass surface to solution volume ratio (S/V). The first leaching experiment was performed with two renewals of ultra-pure water per day during 24 days, yielding a gel called ‘HR gel’ (high renewal rate) ($S/V = 18 \text{ cm}^{-1}$). The second

sample designed as ‘LR gel’ (low renewal rate) was obtained with renewals twice a week during 171 days, ($S/V = 46 \text{ cm}^{-1}$). This second mode is more realistic and representative of geologic storage conditions. Gel compositions have been deduced from the leachates ICP-MS analysis and are given in Table 1.

Site selective spectroscopy was performed with a He circulation cryostat at 14 K under a dye laser excitation (Rhodamine) pumped with a XeCl excimer laser. The luminescence was analyzed with a 1-m monochromator with a 1200 groove/mm grating and recorded by a fast response AsGa photomultiplier. Time-resolved spectroscopy was performed using an intensified CCD camera coupled with a monochromator with a 1200 grooves/mm grating associated with a delay generator.

Mössbauer absorption spectra were obtained in standard transmission geometry, using a source of $^{151}\text{SmF}_3$ with activity 3.7 GBq. Calibration was performed using a source of ^{57}Co in rhodium and a metallic iron foil (25 μm thick) as the absorber. The isomer shift of the samples was measured using the anhydrous fluoride EuF_3 as reference material. Measurements were carried out at room temperature on powder samples with an absorber superficial density of 2.84 mg/cm^2 of Eu for the nuclear glass, 1.35 mg/cm^2 for the HR gel and 1.62 mg/cm^2 for the LR gel. The powders were contained in a Plexiglas holder. Absorption spectra were analyzed by fitting data with multiplet components of Lorentzian shape, allowing for the quadrupole interaction. Analyses were made using the method for pure quadrupole spectra proposed by Shenoy and Dunlap [10], with a value of the quadrupole ratio $R = 1.312$ [11].

^{27}Al MAS NMR spectra of powder samples were acquired on a Bruker AVANCE 500 spectrometer operating at a Larmor frequency of 130.06 MHz. A Bruker high-speed MAS probe with 4-mm rotor was used and sample spinning speed was 12.5 kHz. Up to 512 Free Induction Decays were accumulated with single pulse excitation of length 1 μs and recycle delay of 1 s. ^{27}Al NMR frequencies were externally calibrated against a 1M $\text{Al}(\text{Cl})_3$ solution.

Electronic microprobe (EM) measurements were performed in B. Pascal University (Clermont-

Table 1
Glass LR and HR gels compositions given in elemental wt%

Elements	Glass	LR gel	HR gel
Si	22.82	21.51	18.76
Al	2.62	6	5.76
B	4.68	0.06	0
Zr	1.87	5.67	6.84
Na	3.87	1.77	2.11
Li	0.88	0.16	0.2
Cs	1.17	0.22	0.26
Ca	2.84	3.43	2.46
Sr	0.29	0.34	0.25
Cr	0.33	0.99	0.72
Ba	1.55	0.66	0.47
Te	0.14	0	0
P	0.13	0.37	0.51
Y	0.15	0.45	0.54
Zn	2.01	6.08	7.33
Ni	0.58	1.76	2.12
Eu	0.81	2.7	3.56
La	2.54	8.5	11.17
Co	0.09	0.28	0.35
O	50.63	39.05	36.54

The glass composition is obtained from ICP-MS analyses and the gels compositions are obtained from calculations based on the element amount present on the leachates.

Ferrand) with a Cameca SX 100. Elemental maps were realized on gel pellets of 1 mm of diameter.

Particle induced gamma ray emission measurements were realized in P. Sue Laboratory (CEA Saclay) with incident protons of 3.5 MeV on gel pellets of 1 mm of diameter.

3. Results

3.1. Site selective spectroscopy

3.1.1. Emission spectra

$^5D_0 \rightarrow ^7F_1$ emission spectra were recorded at 12 K under a tunable dye laser excitation with a 0.5 nm step between 576 and 580 nm. In general case, 7F_1 level is decomposed in 3 Stark components. In the present case, each emission band for glass and gels is deconvoluted in 6 Voigt components as shown on Fig. 1 for LR gel. These 6 components are assigned to the existence of two site distributions, after having checked that energy transfer processes were not responsible for them. Experimental conditions (temperature, Eu concentration) were indeed optimized in order to avoid energy transfers [5]. As a matter of fact, time-resolved emission spectra remain unchanged when applied gate and delay are modified. The 7F_1 component is decomposed into 6 levels corresponding to two sites called I and II multiplied by three Stark levels:

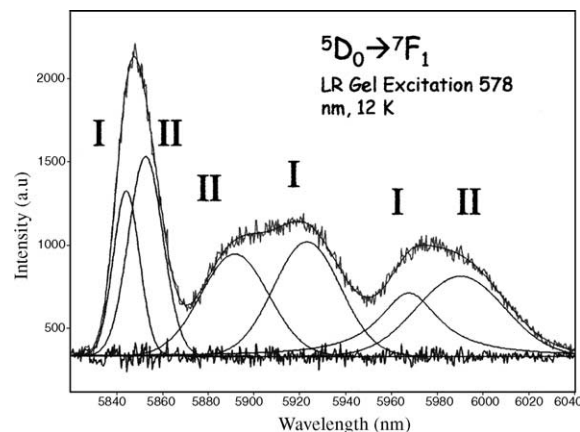


Fig. 1. $^5D_0 \rightarrow ^7F_1$ emission spectrum of Eu^{3+} in LR gel under a 578 nm excitation at 12 K.

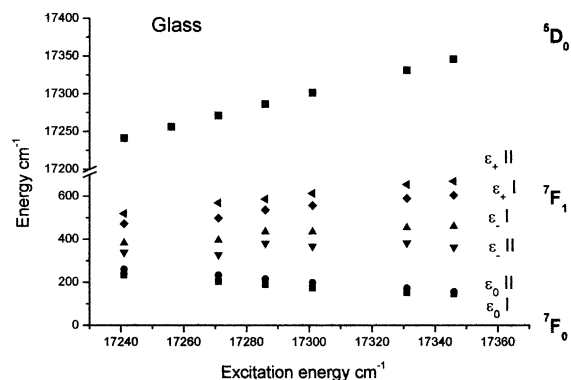


Fig. 2. 7F_0 , 7F_1 , 5D_0 , Eu^{3+} energy levels as a function of the excitation energy.

ϵ_0 , ϵ_- and ϵ_+ with increasing energy. 7F_0 , 7F_1 and 5D_0 glass energy levels are represented on Fig. 2.

The crystal field potential H_{cf} acting on Eu^{3+} can be expressed by the following equation: $H_{cf} = \sum_{k,q} B_{kq} C_{kq}$ where the B_{kq} are the crystal field parameters and C_{kq} spherical harmonics. Considering the C_{2v} symmetry according to Brecher and Riseberg [6] and the experimental energy levels determined from FLN spectra, the crystal field parameters were calculated ignoring the J mixing according to the equations given by Pucker et al. [5]. For each sample (glass, LR and HR gels) B_2 , B_{20} and B_{22} crystal field parameters from sites II and I are listed in Table 2.

Crystal field parameters can be useful to identify the Eu^{3+} site nature; some examples are given in the literature [5,6]. On Fig. 3, LR, HR and glass B_2 parameters from site I are plotted as a function of the excitation energy and compared to literature. From this figure, we can see that B_2 parameters from glass site I suit on well with B_2 parameters attributed to a silicate site.

3.1.2. Excitation spectra

$^7F_0 \rightarrow ^5D_0$ excitation spectra of Eu^{3+} in gels and in the glass were recorded at 12 K. The 587 nm emission is characteristic of the site II (see Fig. 1). The excitation spectrum of this emission with a 0.6 nm resolution is displayed on Fig. 4. The full width at half maximum (FWHM) of the glass and of the LR gel excitation spectra is relatively narrow (30 cm^{-1}) and reflects the precision of the site selection.

Table 2

B₂ parameters in cm⁻¹ of glass (Panel A), of site I for LR and HR gels (Panel B), of site II for LR and HR gels (Panel C)

Excitation (nm)	Glass site I			Glass site II					
	B ₂₀ (cm ⁻¹)	B ₂₂ (cm ⁻¹)	B ₂ (cm ⁻¹)	B ₂₀ (cm ⁻¹)	B ₂₂ (cm ⁻¹)	B ₂ (cm ⁻¹)			
<i>Panel A</i>									
576.5	-643	362	822	-600	768	1241			
577	-616	340	781	-575	681	1122			
577.5									
578	-534	305	687	-485	617	998			
578.5	-485	276	623	-436	575	922			
579	-407	256	544	-359	604	926			
579.5									
580	-322	220	447	-280	454	700			
	HR gel site I			LR gel site I					
<i>Panel B</i>									
576.5	-621	327	774						
577	-575	320	731	-574	362	770			
577.5	-535	319	700	-533	327	705			
578	-467	334	664	-489	327	673			
578.5	-457	305	628	-438	305	615			
579	-382	270	540	-377	284	551			
579.5	-317	192	417	-339	256	496			
580	-271	199	391						
	LR gel sitel I			HR gel site II			From [19]		
<i>Panel C</i>									
575.7								1242	
576.5				-544	776	1225	-1250	551	1141
576.9							-1183	510	1064
577	-502	825	1270	-505	776	1208			
577.3							-1113	474	991
577.5	-478	768	1186	-480	774	1196			
577.6							-1017	429	900
578	-434	739	1132	-417	746	1134			
578.1							-970	400	846
578.5	-376	654	999	-414	681	1048			
578.9							-768	320	675
579	-300	555	840	-293	463	717			
579.5	-280	483	739	-261	426	657			
579.7							-617	282	577
580				-237	398	610			

The observed shift between the glass and LR gel indicates an evolution of site II with leaching while the excitation spectrum of HR gel is the same as LR gel.

3.2. General photoluminescence results

Fig. 5 represents the Eu³⁺ emission spectra at 300 K of the glass and of the LR and HR gels

under a 578 nm excitation; ⁵D₀–⁷F₁ and ⁵D₀–⁷F₂ transitions are observed. The asymmetry ratio (As) defined as the intensity ratio between the ⁵D₀ → ⁷F₂ and the ⁵D₀ → ⁷F₁ emissions, represents an interesting parameter revealing the centrosymmetry distortion of the Eu³⁺ site. The As value for the glass is close to 3, suggesting a low symmetry site for Eu³⁺ in agreement with usual

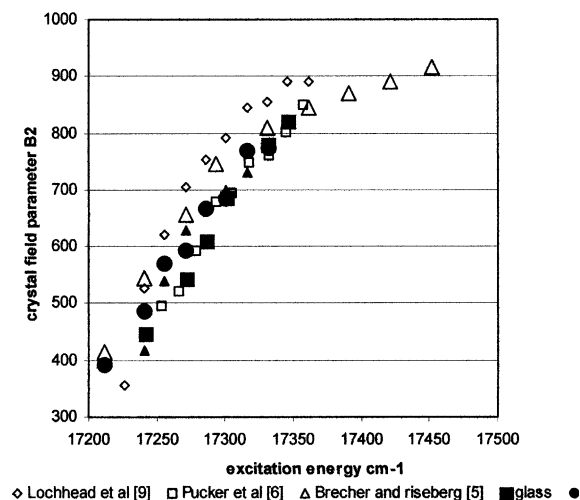


Fig. 3. Comparison of B_2 values for the site I from the literature (open symbols) and this work (closed symbols).

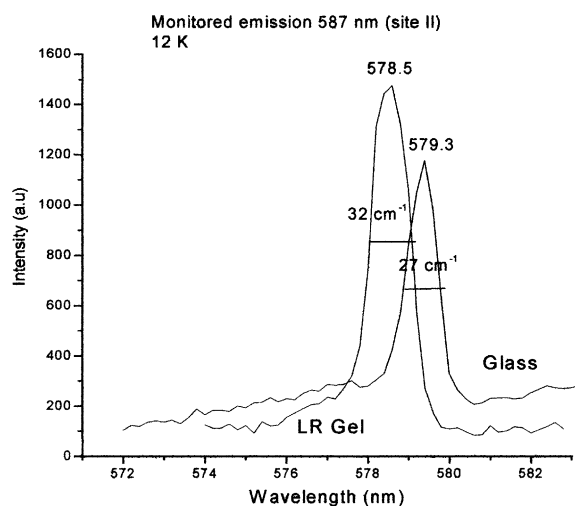


Fig. 4. ${}^7F_0 \rightarrow {}^5D_0$ excitation spectra of Eu^{3+} in gels for a 587 nm monitoring emission at 12 K.

reported values between 3 and 4.25 for borosilicate glasses [7]. LR and HR gels present lower values close to 2 that may be interpreted in term of a decrease of Eu–O covalency and/or a decrease of the average distortion of the Eu^{3+} sites. Indeed, the ${}^5D_0 \rightarrow {}^7F_2$ is an electric dipole transition, hypersensitive and its intensity increases with increasing ligand covalence or increasing site distortion. On

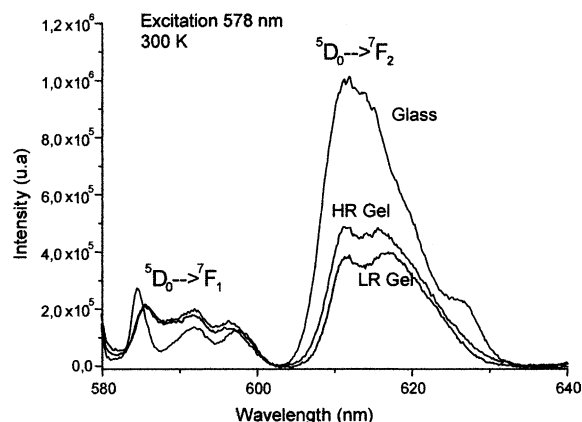


Fig. 5. 300 K Eu^{3+} luminescence spectra of glass, LR and HR gels under 578 nm excitation in the 575–640 nm range.

the contrary, ${}^5D_0 \rightarrow {}^7F_1$ being an allowed magnetic transition is much less sensitive to the Eu^{3+} environment [7]. The decrease of the Eu^{3+} site distortion is in agreement with the non-exponential decay time measurements (Fig. 6). They show that the long component close to the 5D_0 lifetime is longer for gels (1.8 ms) than for glass (650 μs).

3.3. Mössbauer spectroscopy

The experimental absorption spectrum of the glass is displayed on Fig. 7(a) and the spectrum of the LR gel on Fig. 7(b) with their best fits. The

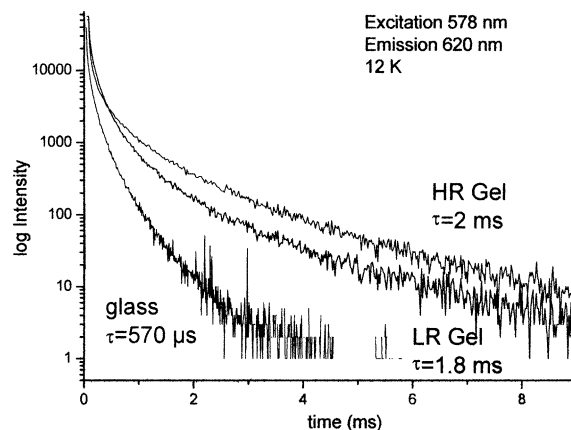


Fig. 6. 5D_0 lifetime decay of Eu^{3+} in glass, LR, HR gels recorded at 12 K (excitation 578 nm, emission 620 nm).

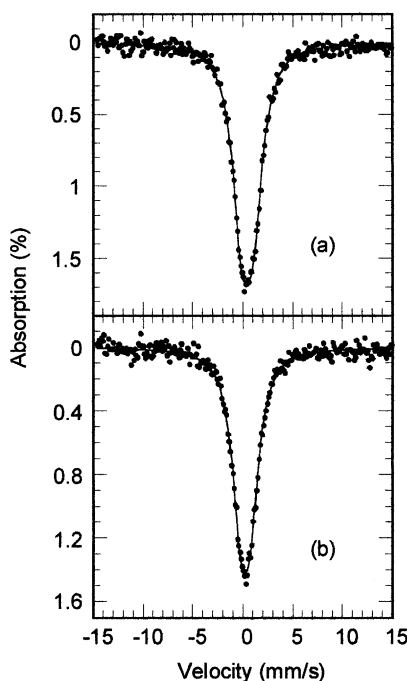


Fig. 7. Absorption spectra of: (a) nuclear glass and (b) LR gel. The experimental data (dots) and the fit (full line) are shown.

parameters obtained by fitting the data of the three samples are given in Table 3. The samples show differences with physical sense (larger than the experimental error) in the values of isomer shift and quadrupolar interaction parameter.

All the samples have isomer shift corresponding to an oxidation number +3; the values are about in the middle between the most ionic compounds (~ 0 mm/s) and the most covalent compounds

(~ 1 mm/s) reported in the literature [12]. In particular the isomer shift is smaller in the HR and LR samples than in the glass; it can be explained as a decrease of the average covalence of the Eu–O bonds in the two gels with respect to the glass.

The quadrupolar interaction parameter is related to the distortion of the Eu^{3+} sites; values between -5 and -8 mm/s are reported for most of the trivalent europium oxides [12]. The decrease of the absolute value of the quadrupolar parameter in the HR and LR samples shows that the Eu^{3+} sites are, on average, less distorted in the two gels than in the glass.

4. Discussion

FLN experiments carried out on the glass and both gels enabled us to detect two different environments for Eu^{3+} . In order to identify them more precisely, B_{20} , B_{22} and B_2 crystal field parameters were calculated. For site I, no ambiguity was found. B_2 parameters from literature associated to a silicate site for different glassy matrix: a Na–Ba–Zn silicate [5], a borosilicate [6] and a Na disilicate [9] are in agreement with our values for site I. It demonstrates that the site I for glass as well as for LR and HR gels can be attributed to a silicate environment (second neighbours). This site moreover remains unchanged after leaching in both LR and HR gels.

Concerning the second Eu^{3+} environment, in glass, a boron site is strongly suspected. First, because the B_2 glass values are in good agreement with those of a borate site as calculated Pucker et al. [6] and secondly, by analogy with the recent results concerning Nd^{3+} . It was recently shown in aluminoborosilicate [13] and borosilicate glasses [14] that Nd ions have a strong affinity with non-bridging oxygen attached to boron atoms. Li et al. [13] showed that the local structure of Nd–O in glasses can be compared to lanthanum metaborate i.e., Nd forming bonding with its surrounding oxygens from two trigonal boron and one tetrahedral boron where Al_{IV} substitutes for B_{IV} . These results are likely not limited to the Nd^{3+} environment and can be extended to Eu^{3+} ions in agreement with our experimental results.

Table 3
Mössbauer parameters as obtained by fitting the spectra of the glass, LR and HR gels

Samples	δ (mm/s)	Γ (mm/s)	eQV_{zz} (mm/s)	η
Glass	0.66 (1)	1.86 (4)	-6.8 (2)	0.87 (5)
HR gel	0.45 (1)	1.81 (5)	-5.6 (2)	0.86 (8)
LR gel	0.44 (1)	1.84 (6)	-5.6 (3)	0.92 (9)

δ is the isomer shift with respect to anhydrous fluoride (EuF_3); Γ is the FWHM of the Lorentzian components (resulting by the quadrupole splitting); eQV_{zz} is the quadrupole interaction parameter; η is the asymmetry parameter. Errors on the last digit are given in round brackets.

The site II in gels cannot correspond to a borate one because boron is absent. In the LR gel, amount of boron measured by Micro-PIGE was less than 500 ppm (corresponding to the detection limit). The comparison of ${}^7\text{F}_0 \rightarrow {}^5\text{D}_0$ excitation spectra (Fig. 4) shows that site II is modified during leaching. Thus if we consider a reverse reasoning, excitation spectra change indirectly confirms the hypothesis of a boron site in glass. The diminution of the covalent character of the average Eu–O bond showed by Mössbauer spectroscopy in gel is an other argument to assess the site nature evolution after leaching.

In gels, aluminium is the stronger former after silicon. The hypothesis of an aluminate site is inferred by recent papers mentioning the existence of a REE environment constituted by Al atoms [15–17]. EXAFS studies showed the existence of Nd–O–Si/Al bonds in aluminosilicate matrix [15]. The same idea was suggested for Er^{3+} in xerogel [16] and a study on Sm in aluminosilicate glasses reported that Sm^{3+} is surrounded by AlO_4 units [17]. In all these cases, the lanthanide plays a role of a charge balancing cation. In nuclear glasses, Al^{3+} is organized in AlO_4 species, Na^+ and preferentially Ca^{2+} insuring the charge neutrality [18]. In gel, AlO_4 species are dominant [19]; however ${}^{27}\text{Al}$ MAS NMR spectrum performed on the LR gel (Fig. 8) shows that a little part of aluminium in sixfold coordination is created with leaching. We compared B_2 parameters obtained in gels with those of an AlO_6 site described by Nogami et al. [20]. The values are close (see Table 2) and could infer to REE a role of charge balancing cation. To confirm this hypothesis, we tried to realize global charges estimation in LR gel from the theoretical composition given in Table 1. The necessary charges to compensate ZrO_6^{2-} , AlO_4^{2-} and ZnO_4^{2-} entities represent 13.4 negative charges. Alkali ions and Ca^{2+} deliver about 7.4 positive charges, non-sufficient to compensate all the entities in gel. Thus, this simple calculation suggests that the role charge balancing cation could be played by REE. As a matter of fact, the number positive charges brought by REE equals 6.

Concerning the evolution of Eu^{3+} sites, the idea of a higher order around the lanthanide was demonstrated by the decrease of asymmetry ratio

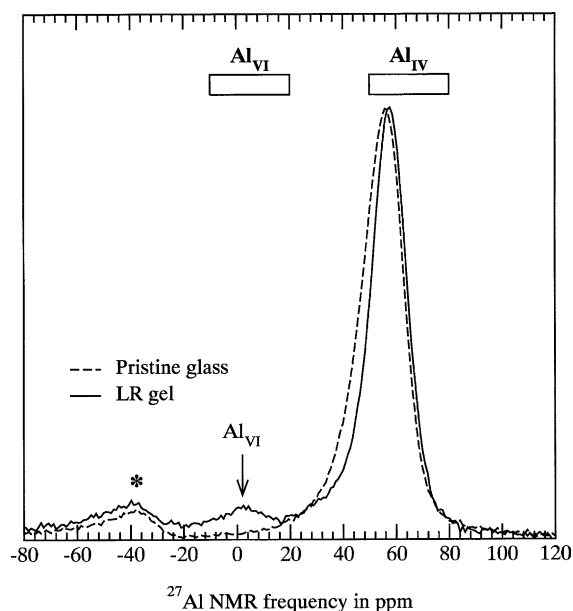


Fig. 8. ${}^{27}\text{Al}$ MAS NMR spectra of the pristine glass (dashed line) and the LR gel (solid line). Spinning sidebands are marked by star (*). Usually observed chemical shift range for four-fold and six-fold coordinated aluminium in silicate are also indicated. The asymmetric lineshape of the glass peak results from a distribution of the quadrupolar interaction.

corroborated by Mössbauer results detailed in 3.3. This decrease of disorder in gel is inferred by the following complementary results: Eu^{3+} site distribution in glass and gels was compared by recording ${}^7\text{F}_0 \rightarrow {}^5\text{D}_0$ excitation spectra (the ${}^5\text{D}_0 \rightarrow {}^7\text{F}_2$ emission at 620 nm is monitored). The FWHM is 60 cm^{-1} in the case of LR and HR gels against 85 cm^{-1} for glass. This result clearly shows that in HR and LR gels the Eu^{3+} site distribution is narrower than in glass. The idea of a more organized structure is coherent with an earlier study [20] reporting that the important diminution the 1000 cm^{-1} IR vibration band FWHM in gel traduces a more regular silicon network. Moreover, one may notice the decrease of the width of the NMR Al line from the glass to the gel (Fig. 8). It mainly results from a decrease of the width of the quadrupolar interaction distribution, thus revealing a gel structure less disordered than the pristine glass structure.

All these results attesting an order increase around Eu^{3+} could also be interpreted as a beginning

of a crystallization including europium. However, europium elemental map realized by electronic microprobe revealed that the element repartition in gel is quite homogeneous; we only observe some phosphorus precipitations associated to La [21]. X-ray diffraction measurements confirm the amorphous character of LR and HR gels.

A last important result to be pointed out is that the leaching mode (renewal rate and duration) in the studied range does not influence the rare earth behaviour. As a matter of fact, LR and HR gels gave strictly the same spectroscopic results. This feature means that the rare earth evolve very quickly towards its new environment and the rearrangement is probably made in situ. The LR gel has a final dissolution rate six times as small as the HR gel dissolution rate. In kinetic terms, the LR can be considered more protective than the HR gel [22]. So the last result from this study is that the protective character of gel does not seem to be linked to the rare earth environment.

5. Conclusions

The local structure of Eu^{3+} in a nuclear glass as well as in leached gels was described using site selective spectroscopy. Two Eu^{3+} environment types were found in the glass as well as in the gels and they could be identified by studying crystal field parameters (B_2). For both gels and glass, the first site was attributed to a silicate site. The second one is probably constituted of boron atoms in the case of glass and of Al atoms in sixfold and fourthfold environment in the case of gels. Glass and gels are both amorphous materials but gel structure is relatively different from glass. The major differences between them lie in the site distribution, which is narrower for LR and HR gels and the average distortion of the Eu^{3+} sites, which is smaller in gels. No trace of crystallization including Eu was observed. In conclusion, this work shows that LR and HR gels present no fundamental difference and the leaching mode in the studied range does not seem to influence Eu behaviour. Moreover, the protective character of gel

is likely not dependant on structural feature linked to the rare earth element environment.

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